Metadata management, publishing and discovery using OGC API - Records

Tom Kralidis
Senior Geospatial Architect, WMO WIS2
Chair, WMO Expert Team on Metadata Standards
Co-Chair, OGC API – Records Standards Working Group
Board of Directors, Open Source Geospatial Foundation

27 January 2026

Agenda

- OGC API – Records recap
- Tooling overview
  - pygeometa
  - pygeoapi
  - QGIS

It’s all about metadata!

- Description/documentation
- Identification
- Contact(s)
- Access
- Fees and usage

OGC API - Records

- Standard that provides discovery, query and modify capability for geospatial metadata about any resource on the Web
  - APIs
  - Content models
- Part 1: Core
  - Published 2025
  - ogcapi.ogc.org/records
  - Specification: docs.ogc.org/is/20-004r1/20-004r1.html
- Part 2: Facets (in development)
- Part 3: Create, Replace, Update, Delete, Harvest (in development)

Implementations/Adoption: WMO WIS2

- UN/WMO Information System (WIS2)
  - next generation international data exchange for all Earth System Data
    - weather, climate, water
  - 193 countries
  - Discovery, access, retrieval
  - Replaces WMO legacy systems (Global Telecommunication System [GTS])

WORLD
METEOROLOGICAL
ORGANIZATION

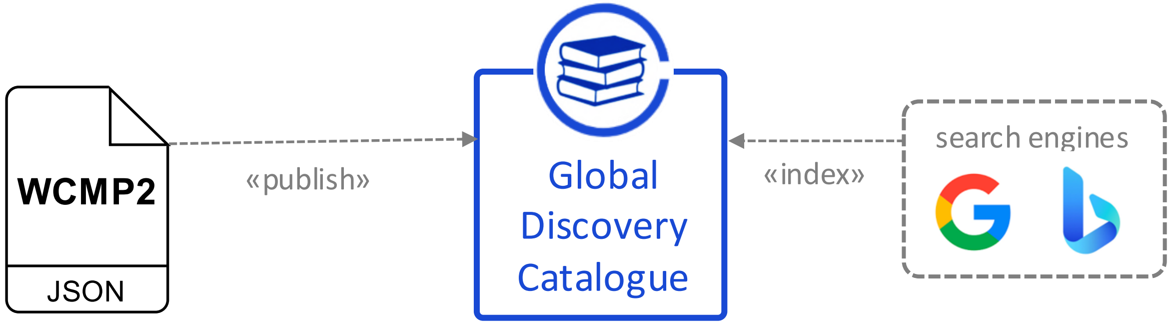
WIS 2.0

… collaborative system of systems using Web-architecture and open standards to provide simple, timely and seamless sharing of trusted data and information …

- Open Standards (OGC, W3C, IETF, …)
- Free and Open Source tooling
- Data sharing through Web and real-time notifications with publication/subscription (Pub/Sub) protocols
- Cloud ready (turn-key solutions)
- Web APIs (Application Programming Interface)

WMO Core Metadata Profile (WCMP2)

- WMO Core Metadata Profile 2.0 (WCMP2)
  - extension of OGC API - Records - Record Core
  - discovery metadata describing a given dataset or collection
  - aligning with the WIS 2.0 Principles, discovery metadata published to the Global Discovery Catalogue



Manual on WIS (WMO No. 1060), Vol II, Appendix F: WMO Core Metadata Profile 2 [approved]

Guide to WIS (WMO No. 1061), Vol II, §1.3.2 How to provide discovery metadata to WIS2 [approved]

Description of the Dataset
Identifier Title Description Keywords Geometry (extent) Time (extent)

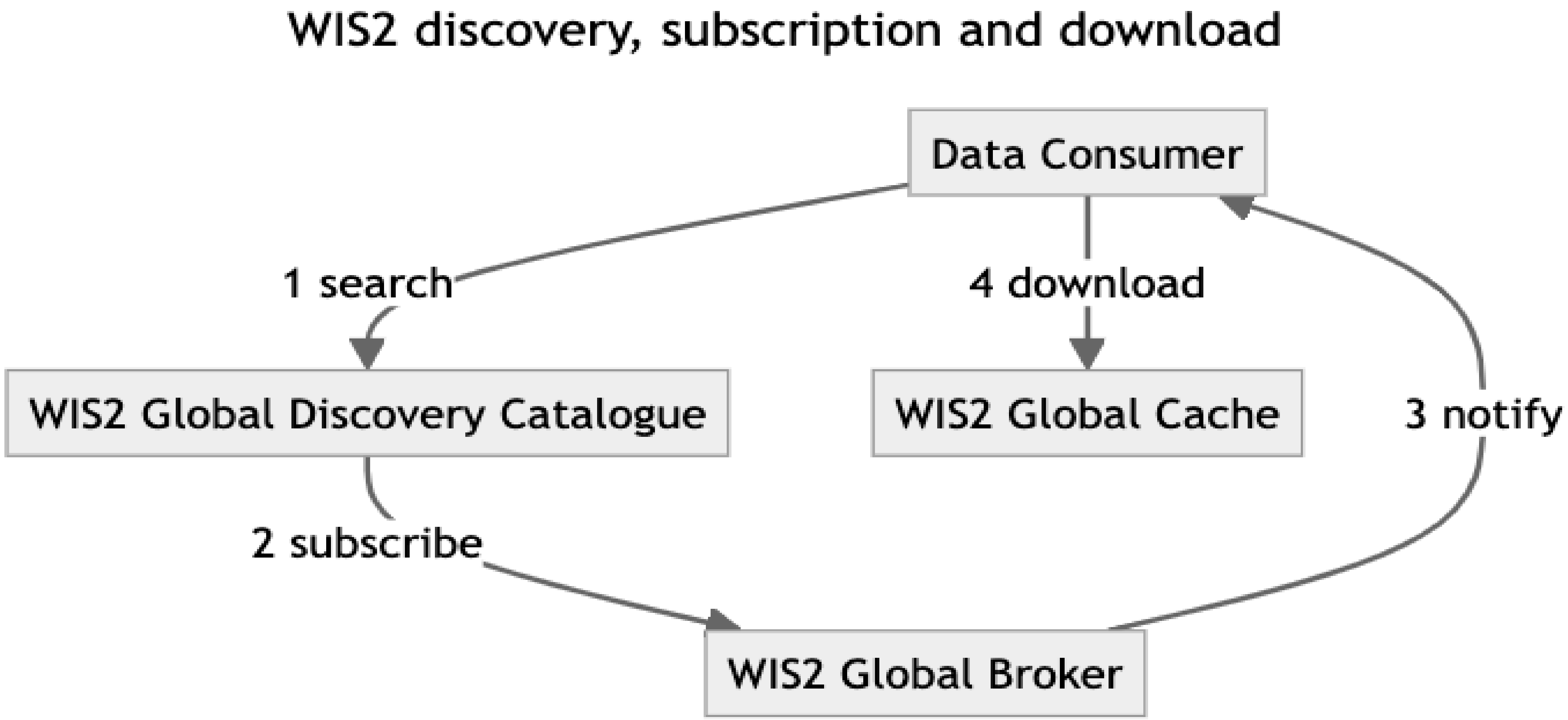
Who to contact
Publisher contact

How to access
Data access (files) Data access (API) Data access (notifications)

Conditions of use
Data policy Rights License

How to attribute
Citation

Event driven: WMO WIS2 data consumer workflow



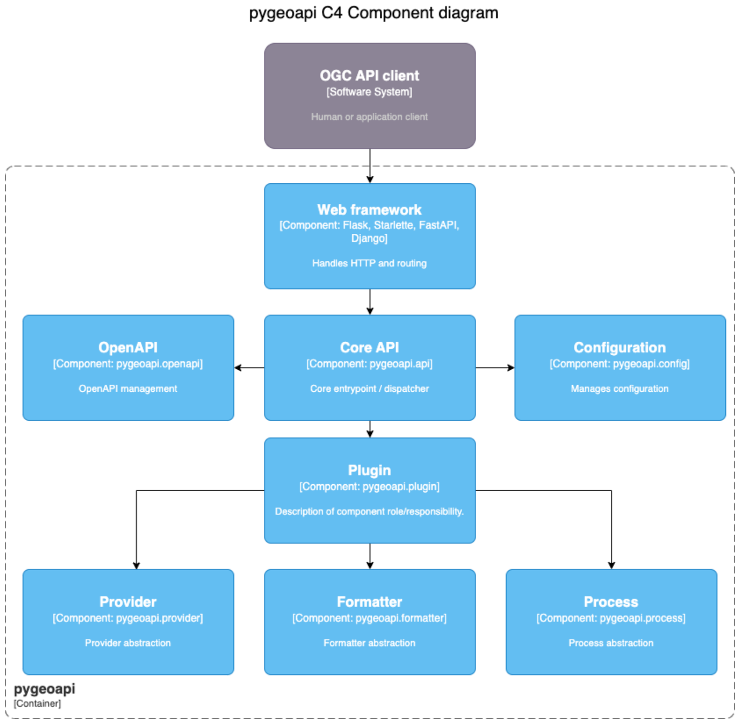
pygeometa

- GDAL-like tooling for geospatial metadata
- Read-write for numerous geospatial metadata formats
- CLI or Python API
- Focus on automated workflow / runtime pipelines
- Multilingual Support
- Core model (‘Metadata Control File’ MCF) -- YAML
    - Support for ‘base’ MCF reuse (i.e. contacts)
- Schema validation
- Extensible (plugins)
    - allows for additional properties
    - add your own Metadata Formats
    - geopython.github.io/pygeometa

OSGeo Community

pygeoapi

- Reference Implementation for numerous OGC API standards
  - Features, Records, Coverages, EDR, Maps, Tiles, Processes, Pub/Sub
- Simple Python server
- Extensible [architecture] ([plugins])
- Dozens of high profile deployments
  - Meteorological Service of Canada
  - US Geoplatform.gov
  - UN/WMO (wis2box)
- Over 100 contributors
- pygeoapi.io

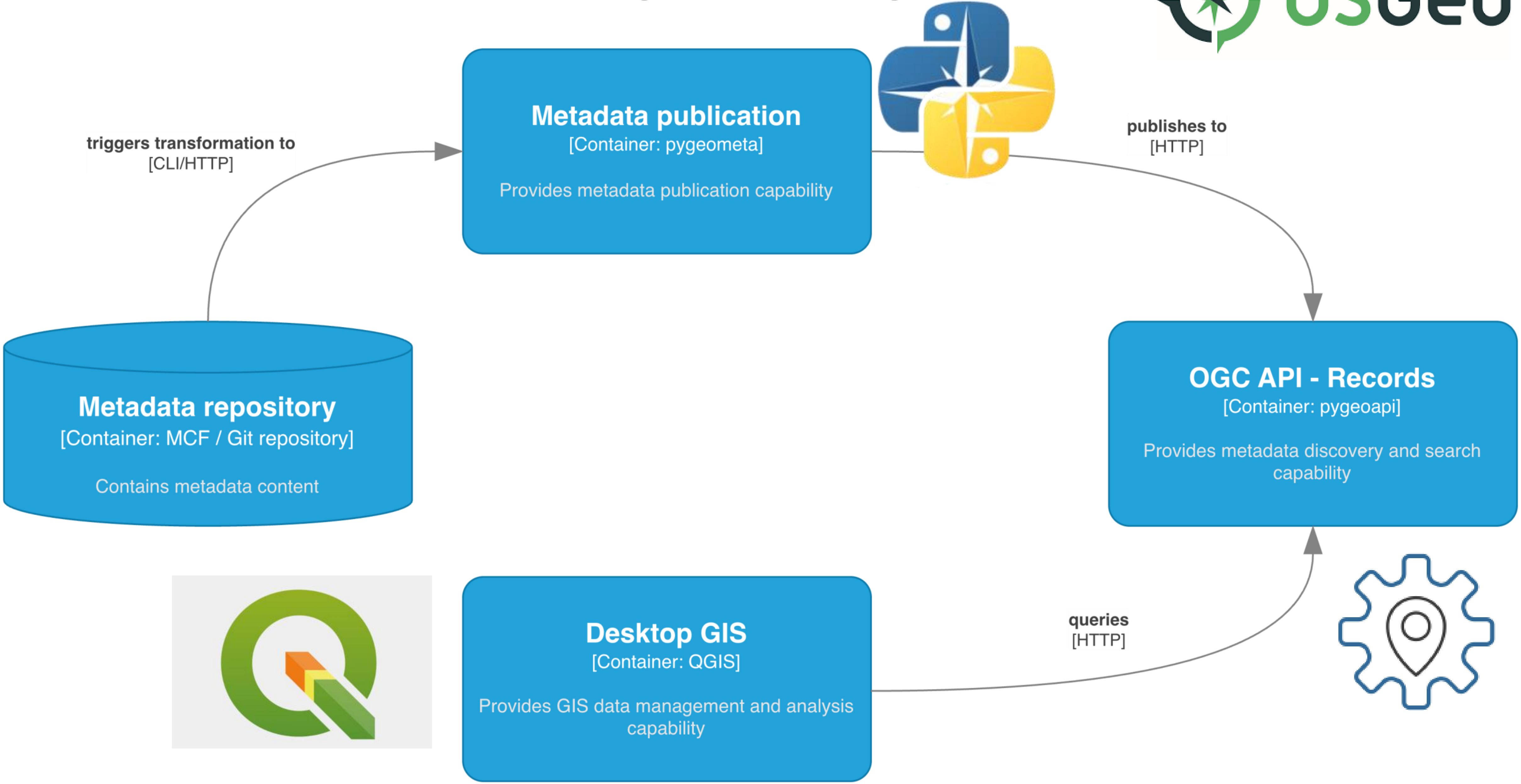


QGIS

- Desktop GIS application
- Cartography/maps
- GIS data management and analysis
- Extensible: over 2000 plugins
- Strong, longstanding support of OGC standards
  - WxS
  - OGC APIs

OSGeo Project

Putting it all together



Thank you

@tomkralidis/noc.social

github.com/tomkralidis

tomkralidis@gmail.com